

THE AI TOOLBOX · CLASS 04

ChatGPT for Research

Turn an LLM into a real research assistant — and learn to **distrust it productively**.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

The pipeline

Data → **Model** → **Experiment** → **Conclusion**
— one loop, every study, and it cycles.

The three piles

Train / validation / test. The test set is sacred:
touch it **once**. Overfitting hides in the train-vs-
test gap.

Your map

You drew your question's pipeline diagram —
with ▲ on the predicted hard part.

Homework share: hold up your pipeline diagram — where did you put the ▲? And who ran their first Colab lap? Today's tool accelerates **every box** on that diagram — which is exactly why it needs a seatbelt.

Where we're going

- **0–8'** Homework share + the lawyer who trusted the chatbot
- **8–20'** Five hats an LLM wears · the anatomy of a research prompt
- **20–30'** Three prompt upgrades: specific · role · example
- **30–40'** The paper-summary framework · hallucinations & the verify routine
- **40–48'** Live demos: ChatGPT · Claude · the 2-minute check
- **48–54'** Two labs: build a tiny "language model" · score your prompts
- **54–60'** Your verified paper summary, homework & wrap-up
- Quizzes &  comments throughout

By the end: write prompts that get research-grade output, summarize any paper into **summary / contribution / limitation**, and catch a hallucinated claim before it catches you.

WARM-UP · ACTIVATE

Quick check · tap an answer

An AI assistant tells you: 'A 2021 study by Smith et al. proved exactly this.' What's the researcher's move?

Cite it — the AI sounds confident and gave a year

Ask the AI if it's sure (it will say yes)

Search for the study yourself; if you can't open it, it doesn't exist for your purposes

Assume it's fake and never use AI again

THE STORY

The lawyer who trusted the chatbot

In 2023, a New York lawyer filed a court brief citing **six perfect-looking precedent cases** — names, quotes, citations, all immaculate. One problem: **ChatGPT had invented all six**. They did not exist. The judge sanctioned him, and the story went around the world as a warning. The AI wasn't lying on purpose — it was doing exactly what LLMs do: **predicting plausible text**.



Speak this story

Neural voice · click to play / pause

You met this story once before — Class 1's Perplexity demo asked you to fact-check it. Today you learn the machinery that produced those six fake cases, and the 2-minute routine that would have caught them.

One tool, five hats

"Chatbot" undersells it. An LLM is a **text-processing engine** that plays whatever role you assign — and each role helps a different pipeline stage:

Researcher

Summarizes papers, explains concepts in plain words. (*Question & data stages.*)

Programmer

Writes, explains, and debugs code — you've used this one: the labs' ✨ assistant. (*Model stage.*)

Data analyst

Suggests analyses, writes chart code, interprets results. (*Data & experiment stages.*)

Mentor

Critiques your plan, spots gaps, suggests directions when you're stuck. (*Every stage.*)

Writing partner

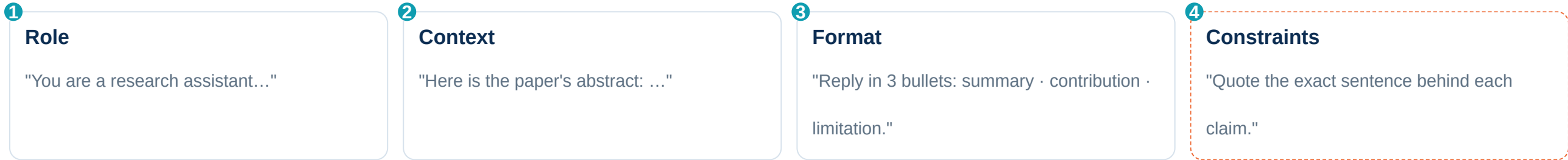
Drafts outlines and abstracts, polishes academic prose. (*Conclusion stage.*)

You pick the hat

The role doesn't switch itself — **your prompt assigns it**. That's the skill we learn next.

Anatomy of a research prompt

Vague in, vague out. Four parts turn a wish into an instruction:



Result ▶ a useful, **checkable** answer – the quoted sentences are your verification handles.

KEY IDEA — a good prompt names the **exact format** you want the answer in. The constraints part is dashed for a reason: it's the piece almost everyone forgets, and it's the piece that makes answers verifiable.

PROMPT UPGRADE 1 OF 3

Vague → specific

Vague

You ▶ Explain AI.

No audience, no scope, no format → a generic essay that helps nobody.

Specific

You ▶ Explain artificial intelligence to a 10-year-old, using two everyday analogies, in under 100 words.

Audience + format + length → a targeted, vivid, usable answer.

The move: answer *who, what, why, how* inside your prompt. Sound familiar? It's the question ladder from Class 2 — vague → specific — **applied to prompts**. Same skill, new target.

Assign the hat: role-playing

No role

You ▶ How do I write a linear-regression function in Python?

You'll get bare code — no reasoning, no guidance, no warnings.

With a role

You ▶ Act as a patient data-science tutor. Teach a beginner to write linear regression with NumPy only: explain the idea first, then the code with line comments, then one common mistake.

Now it **teaches**: background, commented code, pitfalls — like a tutor, not a search engine.

The move: start with "Act as..." or "You are...". The role sets depth, tone, and what the model volunteers unasked — that's the difference between passive Q&A and proactive teaching.

Show, don't describe: give an example

✗ Describing the style

You ▶ Summarize this paragraph in a creative way.

"Creative" means a thousand things — you'll get a lottery ticket: maybe a poem, maybe a story, rarely what you pictured.

✓ Showing the style

You ▶ Summarize it as a haiku. Example — Input: "The quick brown fox..." Output: "Fox leaps, dog dreams on, / a chase painted in the air, / nature's brief ballet."

One example anchors format, tone, and length in a single shot.

The move: when you need a specific format or style, paste **one worked example**. (The pros call this "few-shot prompting" — you just learned it.) Your homework's 3-part paper summary? Give it one example summary and watch the consistency jump.

Quick check · tap an answer

You want an LLM's answer to be easy to fact-check afterwards. Which prompt part does that job, and what would it say?

Role — 'You are a very honest assistant'

Context — paste more of the paper

Format — 'answer in bullets'

Constraints — 'quote the exact sentence behind each claim'

Three things matter about any paper



Summary

What they did, in plain words. One or two sentences a classmate would understand.



Contribution

What's actually new. Not "they studied X" — what can we do now that we couldn't before?



Limitation

Where it falls short — the scenario not covered, the data not tested. **This is where your project ideas live.**

You ▶ You are a research assistant. Here is a paper's abstract: [...]. Reply in exactly 3 bullets – summary, contribution, limitation – and quote the sentence that supports each.

Remember Class 2's secret? "Future work" sections are free research questions. A paper's **limitation** is the same gold from the other side — and now an LLM mines it for you in seconds. *Then you verify.*

THE MAP · ONE ASSISTANT, EVERY STAGE

Where it helps across your pipeline



Literature & ideas · question stage

"Summarize this paper's method and findings" · "Explain transfer learning with an example" · "Suggest 5 research questions about ____" — hours of reading compressed to minutes **of drafts to verify**.



Code help · model stage

"Write pandas code to load this CSV and average column X" · "Explain this code line by line" · "Why does this throw an error?" — the syntax barrier drops; your energy goes to the research logic.



Data analysis · data & experiment stages

"Suggest 5 questions worth asking of this dataset" · "Matplotlib code for an age histogram" · "Interpret this confusion matrix" — from raw table to insight, faster.



Writing · conclusion stage

"Outline a paper on ____" · "Draft an abstract from these bullet points" · "Make this paragraph more concise and academic" — drafting and polish, end to end.

Every cell = a hat + a pipeline stage + a starter prompt. Steal these verbatim — then upgrade them with role, format, and constraints.

Hallucination: confident nonsense

What it is: the model generates plausible-sounding but false information — fake studies, fake numbers, fake people — in a perfectly confident tone.

Why it happens: an LLM doesn't look facts up — it **predicts the next likely word**, over and over. When the truth isn't strongly in its training data, the most *plausible* continuation wins. Grammar: perfect. Facts: optional.



High-risk zones

- Specific citations & paper titles
- Precise numbers and statistics
- Names you've never heard of
- Anything "too good to be true"

The rule: never cite anything you haven't opened yourself. If you can't find the source, **it doesn't exist for your purposes** — no matter how real it sounds. You'll build this exact mechanism in Lab 1 and see why it can't help itself.

CASE STUDY · A HALLUCINATION IN THE WILD

Meet Dr. Alice Smith (who doesn't exist)

You ▶ Tell me about the 2025 quantum-computing breakthrough by Dr. Alice Smith.

AI ▶ Dr. Alice Smith, a top researcher at MIT, announced a major breakthrough in early 2025. Her team built a 1000-qubit computer, achieving quantum supremacy – solving in 3 hours a problem that would take classical supercomputers 10,000 years. Published in *Nature*, March 2025.

Red flag 1

Hyper-specific details — 1000 qubits, 3 hours, 10,000 years — for an event you can't find anywhere.

Red flag 2

A named researcher + institution + journal — all unverifiable. The *question invited* the invention.

The check

Search "Alice Smith MIT quantum 2025" in Scholar/news. Zero authoritative hits → hallucination. **Time: 90 seconds.**

CHECK · SPOT THE MECHANISM

Quick check · tap an answer

Why does an LLM state fake citations in exactly the same confident tone as real ones?

It's programmed to deceive users

It predicts plausible next words — it has no internal fact-checker distinguishing memory from invention

Its developers didn't pay for the citation database

It's confident only when it's correct, so fakes must be rare

DISCUSS · 5 MINUTES · DESIGN THE SEATBELT

Save the lawyer: a 2-minute checking routine

The lawyer had six citations and a deadline. Design the routine that would have saved him — **concrete steps, under 2 minutes per claim.**

Your routine

Where do you search first? What counts as "verified"? What do you do with a claim you *can't* confirm either way?

Edge cases

When is it OK to skip checking? (Brainstorming? Code that you'll run anyway? Grammar fixes?) Where's the line?

Pair up, 3 minutes — then we'll assemble the class routine. Drop your best step in .

LIVE DEMO 1

ChatGPT — the full research prompt

All four anatomy parts at once, on a real abstract. Watch the format land exactly as ordered.

You ▶ You are a research assistant. Here is a paper's abstract: [paste]. Reply in exactly 3 bullets – summary, contribution, limitation – and quote the sentence that supports each.

You ▶ Now act as a skeptical reviewer: what's the weakest part of this paper's evidence?

You ▶ Rewrite the limitation as a research question in input → output → metric form.

You ▶ Suggest a public dataset I could test that question on.



chatgpt.com

LIVE DEMO 2

Claude — the second opinion

A different LLM with different strengths — long documents and careful reasoning. Two models disagreeing is a research signal.

You ▶ [Paste the SAME abstract + prompt from demo 1] — compare Claude's limitation with ChatGPT's. Same? Different?

You ▶ Upload the full PDF: summarize section by section, then list every quantitative claim with its section number.

You ▶ Here is ChatGPT's summary of this paper: [paste]. What did it miss or overstate?

You ▶ Explain this paper's method to me step by step, assuming I know Class-3-level ML.



claude.ai

LIVE DEMO 3

The 2-minute verify, live

Close the loop: take the quoted sentences from demo 1 and hunt them down in the real world.

You ▶ Google Scholar: paste the paper's exact title – does it exist? Right venue? Right year?

You ▶ Open the PDF (or arXiv page) → Ctrl-F a distinctive phrase from the quoted sentence.

You ▶ Check the claimed contribution against the paper's own abstract – did the LLM overstate it?

You ▶ Verdict per claim: ✓ verified · ✗ wrong · ? can't confirm → treat ? as ✗.



scholar.google.com

Build a tiny 'language model' — and watch it hallucinate

LLMs predict the next word. That's the whole trick — and the whole problem. This 25-line model learns from a tiny corpus, then **generates confident text with zero understanding**. Sound familiar?

```
# A tiny "language model": it ONLY predicts a likely next word.
import random
random.seed(4)

corpus = """the model learns patterns from data . the model predicts
the next word . the data trains the model . the model sounds confident .
the model does not check facts . the lawyer cited six fake cases .
the judge checked the cases . the cases did not exist . the model
predicts plausible text . plausible text is not always true ."""

words = corpus.split()

# learn: for every word, which words follow it? (a "bigram" table)
next_words = {}
```



Run



Explain



Debug



Improve

Python loads on first Run

► Run the code to see the output here.

AI Assistant



Haiku · fastest



Hi! Edit the code and hit **Run**. Stuck? I can **Explain**, **Debug**, or **Improve** it — or just ask me below.

Ask

A prompt-anatomy checker

Class 2 had a question checker; here's its sibling for prompts. It looks for the four anatomy parts — **role**, **context**, **format**, **constraints**. Score the demos, then score **your own**.

```
# Prompt-anatomy checker: does your prompt have the four parts?

ROLE_SIGNS      = ["you are", "act as", "as a "]
CONTEXT_SIGNS   = ["here is", "abstract", "my data", "my plan",
                  "this paper", "this code", "[paste]"]
FORMAT_SIGNS    = ["bullets", "bullet", "table", "3 parts", "format",
                  "summary", "contribution", "limitation", "steps"]
CONSTRAINT_SIGNS = ["quote", "cite", "exact", "under ", "only",
                  "must", "at most", "word", "sentence"]

def check(prompt):
    p = prompt.lower()
    parts, score = [], 0
    for name, signs in [ ("role", ROLE_SIGNS), ("context", CONTEXT_SIGNS),
```



Run



Explain



Debug



Improve

Python loads on first Run

► Run the code to see the output here.

AI Assistant

Haiku · fastest

Hi! Edit the code and hit **Run**. Stuck? I can **Explain**, **Debug**, or **Improve** it — or just ask me below.

Ask about this code...

Ask

YOUR PROJECT

An AI-assisted, human-verified paper summary

Use an LLM to summarize one real paper into the three parts that matter — then **fact-check every part** against the source.

- Pick a paper — ideally one of the 3 abstracts from your Class-2 homework
- Prompt for **summary · contribution · limitation** with quoted evidence
- Verify each part against the actual paper (the 2-minute routine)
- Note one limitation specific enough to become a research question

Deliverable

A one-page structured summary with a ✓ **verified** note on each claim.

Success looks like

Every claim checked against the source — and a limitation stated precisely enough to feed the input → output → metric recipe.

BUILD IT NOW · HANDS-ON

Summarize, verify, mine

1. Paste your paper's abstract into ChatGPT or Claude with the **full anatomy prompt** (steal demo 1's)
2. Run the **2-minute verify** on each of the three parts — Scholar + Ctrl-F the quoted sentences
3. Mark each claim ✓ / × / ? — rewrite anything ×, treat ? as ×
4. Rewrite the limitation as an **input** → **output** → **metric** question
5. Bonus: feed the same abstract to the *other* model — do the limitations agree?

ChatGPT

Claude

Google Scholar

Tip: when the model quotes a sentence, Ctrl-F a distinctive 4-word phrase from it in the PDF. Found = ✓ in seconds. Not found = you just caught a hallucination in the wild — screenshot it for the class.

What a verified summary looks like

Paper: *"YOLO: unified real-time object detection"* (Redmon et al.) — our AV case's ancestor.



Summary ✓

Frames detection as a **single regression** from image to boxes + class probabilities — one network pass instead of a multi-stage pipeline.
✓ matches abstract, sentence 2.



Contribution ✓

Real-time speed (45+ fps) with one unified model — detection fast enough for live video.
✓ verified in paper's §1 and table 1.



Limitation ✓

Struggles with **small objects in groups** (flocks, crowds) — spatial constraints limit nearby detections. ✓ paper's own §2 admits this.

Mined question: "Do crowd-specific training images raise small-pedestrian recall in groups, without hurting overall accuracy?" — input, output, metric, trade-off guard. From paper to project in one hour. (Notice: every ✓ cites where it was checked.)

Today's vocabulary

LLM — large language model: a next-word predictor trained on vast text (ChatGPT, Claude).

Role prompt — "Act as...": assigns the hat, sets depth and tone.

Hallucination — plausible, confident, false output; a natural by-product of next-word prediction.

Prompt engineering — designing the input to get research-grade output: role + context + format + constraints.

Few-shot example — a worked example inside the prompt that anchors format and style.

Cross-validation (of claims) — confirming AI output against independent, authoritative sources before use.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your verified paper summary** (summary · contribution · limitation, each with ✓ and where you checked it) and submit in Google Classroom.
2. **Design 5 prompts for your own project** — one each for: literature, code, data analysis, hypothesis critique, paper outline. Score them with Lab 2's checker; upgrade until all four parts show.
3. **Use the mentor hat once:** paste your research plan into an LLM, ask for three weaknesses — then decide which criticisms you actually accept, and why.

★ Stretch (optional)

Kaggle's "**NLP with Disaster Tweets**": real tweets, real labels — is each tweet about a real disaster? Use your new assistant to explain the starter code. (*Language as data — today's tool, tomorrow's dataset.*)

[kaggle.com · Disaster Tweets](https://kaggle.com/datasets/rodrigoa/nlp-with-disaster-tweets)

Task 3's last clause is the real assignment: **deciding which AI criticisms to accept** is the Skill of Judging from Class 1, upgraded. Questions → 💬 on any slide.

TODAY'S LINE TO REMEMBER

"The goal is to turn data into information, and information into insight."

— Carly Fiorina

An LLM accelerates the first two steps. The **insight** — and the judgment — stay yours.



Speak this

WRAP-UP

Three things to keep

- Structure your prompts: **role** · **context** · **format** · **constraints**
- Every paper in three parts: **summary** · **contribution** · **limitation** — the limitation is your idea mine
- LLMs predict plausible text — so **verify every claim you'll rely on**; unverifiable = unusable

Exit ticket: post in  one limitation you found today that could become a research project.

Next · Class 5 — Python for data, gently: pandas, real tables, first charts. The assistant you trained today will be your co-pilot.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

Skills AI can't replace

your value in the AI era

The mentor dialogue

a worked plan-review example

Bonus quiz

one more check

When AI drafts everything, what's left for you?

? Asking good questions

Defining the problem beats solving it — Class 2's whole lesson. AI answers; **you** decide what's worth asking.



Critical judgment

Spotting weak evidence, catching hallucinations, deciding which AI advice to accept — today's whole lesson.



Creativity & connection

Cross-field leaps, unprecedented hypotheses, new framings — prediction machines interpolate; **you** extrapolate.

Plus the one that can't be delegated at all: **ethical responsibility** — anticipating your research's impact (Class 3's backup). The pattern: AI amplifies execution; humans own **direction, judgment, and responsibility**. Those are exactly the skills this course grades.

The mentor hat, in action

You ▶ Act as a research advisor. My project: predicting student exam scores from study hours and past grades with linear regression. Critique my plan – what am I missing?

AI ▶ Three gaps: (1) **Features** – hours + grades is thin; consider attendance, sleep, extracurriculars. (2) **Baseline & iteration** – start with linear regression as baseline, then test random forest / gradient boosting and compare honestly. (3) **Evaluation** – you haven't said how you'll split data or which error metric decides success.

Accept ✓

Baseline-first and the split critique — straight from Class 3's playbook. It's right.

Accept with care ⚠

More features — but sleep data may be uncollectable. Feasibility is **your** call, not the model's.

Push back ✖

If it suggests deep learning for 200 rows of data — overkill. You know your scope; it doesn't.

Quick check · tap an answer

Your LLM produces a beautiful summary of a paper, with quoted sentences. You can't find one quoted sentence anywhere in the PDF. What do you do?

Keep it — the other quotes checked out, so this one's probably fine

Trust it — the model wouldn't invent a direct quote

Treat that claim as unusable, re-prompt or re-read the section yourself, and double-check its neighbors

Delete the whole summary and start from scratch